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Barriers to health care access for sanitation workers in five municipalities of Bangladesh: an evaluation of Shasthya Nirapotta Scheme using multinomial logistic regression approach

Faria Tasnim^{1*} , Md Zarif Oeishik^{2*} , Nijam Uddin Sarkar³, Md Tahmidul Islam², Hossain Ishrath Adib⁴, M. M. Mamshad⁵ and Hasin Jahan²

Abstract

Background Sanitation workers in Bangladesh face significant social, economic and institutional barriers to access healthcare facilities, despite their significant contributions to public health and urban sanitation. This study evaluates the effectiveness of a health care initiative targeting sanitation workers, the Shasthya Nirapotta Scheme (SNS), by identifying factors that influence its frequency of utilization.

Methods A cross-sectional quantitative study was conducted among 155 sanitation workers in five municipalities using structured questionnaire. Bivariate analyses identified candidate variables associated with SNS utilization frequency, which were subsequently included in a multinomial logistic regression model. Model assumptions were verified, including the absence of multicollinearity ($VIF < 2$) and satisfactory goodness-of-fit.

Results Multiple factors were significantly associated with frequent SNS utilization. Compared with females, male workers were less likely to use services “sometimes” ($AOR = 0.28$, 95% CI: 0.09–0.82, $p = 0.021$). The absence of social discrimination increased the odds of frequent use more than eightfold ($AOR = 8.36$, 95% CI: 1.98–35.15, $p = 0.004$). Perceiving the premium as affordable was strongly predictive of frequent utilization ($AOR = 13.75$, 95% CI: 3.73–50.68, $p < 0.001$). A lack of administrative bottlenecks ($AOR = 24.21$, 95% CI: 5.96–98.27, $p < 0.001$), proximity of health facilities ($AOR = 15.57$, 95% CI: 3.87–62.69, $p < 0.001$), and negligible travel time ($AOR = 9.61$, 95% CI: 2.57–35.88, $p < 0.001$) were also strong predictors of frequent use.

Conclusions The findings reveals that beyond financial constraints, sociocultural stigma, bureaucratic inefficiencies and spatial exclusion critically shape healthcare-seeking behavior. This study highlights the urgent need for integrated

*Correspondence:

Faria Tasnim
fariatasnimce@gmail.com
Md Zarif Oeishik
zarifoishik@gmail.com

Full list of author information is available at the end of the article



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policy interventions that go beyond infrastructure and affordability—emphasizing dignity, accessibility, and institutional inclusion—to ensure equitable healthcare access for sanitation workers.

Keywords Sanitation workers, Healthcare access, Shasthya Nirapotta Scheme, Multinomial logistic regression, Social protection, Social stigma, Quantitative analysis

Introduction

Over the past decade, there has been a significant focus on ensuring safe and equitable sanitation for all in Bangladesh, aligning with the global commitment of Sustainable Development Goal (SDG) 6.2. The transition from the Millennium Development Goals (MDGs) to the SDGs marked a shift from merely ensuring safe containment to emphasizing safe disposal and reuse, bringing increased attention to the critical role played by sanitation workers across the sanitation service chain. Like many South Asian countries, Bangladesh's sanitation sector heavily relies on onsite sanitation systems, therefore, sustaining the relevance of sanitation workers has become an essential component of the service chain [1].

Sanitation workers perform essential tasks such as cleaning latrines, emptying pits, transporting waste, and managing sludge treatment and disposal [2, 3]. In South Asia, the prevalence of onsite sanitation systems has historically preserved these roles, which are often carried out by socially marginalized, low-income and lower-caste communities, with the occupation commonly passed down across generations [4, 5]. Owing to the entrenched caste-based stigma, sanitation workers face severe inequalities, discrimination, and socioeconomic and cultural barriers that limit their access to basic rights and essential social services [6, 7]. Moreover, their work exposes them to various occupational hazards, including toxic waste and harmful gases, increasing their vulnerability to chronic diseases such as hypertension, respiratory disorders, kidney and liver disease [8, 9].

Despite these significant health risks and social challenges, sanitation workers remain largely excluded from targeted social and health protection programs. Similar challenges have been documented across South Asia and Africa. For example, in Kenya, Uganda and Ghana, informal employment and weak public health infrastructure result in the systemic exclusion of sanitation workers from health insurance schemes [10]. In India, limited efforts under schemes such as Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM-JAY) and the Self-Employment Scheme for Rehabilitation of Manual Scavengers (SRMS) provide some support [11, 12], but enrollment and utilization remain low due to informal work and low awareness [13, 14]. In Nepal, while social assistance and contributory social protection exist for formally employed workers, no specific targeted schemes for sanitation workers or their families are available [15].

In Bangladesh, although the constitution guarantees social protection and prohibits discrimination on the basis of religion, caste, sex, or place of birth [16], the health and welfare needs of sanitation workers remain largely neglected in national development agendas. While labor laws mandate workplace safety and health compensation [17, 18], local government institutions (LGIs) are tasked with ensuring that licensed operators provide health check-ups and insurance for formal workers [19]. Unfortunately, most sanitation workers operate informally, excluding them from these protections.

Recently, several initiatives by LGIs, private sectors and international/nongovernmental organizations have aimed to provide equitable healthcare for sanitation workers. Among these, the Shasthya Nirapotta Scheme (SNS) stands out as a pioneering effort and is currently implemented in eight municipalities. SNS offers significant benefits, including up to 40% discounts at 216 hospitals for outpatient and inpatient services, medications, diagnostic tests, and death coverage of BDT 30,000, all for an affordable annual premium of BDT 199 [20].

Since socioeconomic health equity is closely linked with social, economic, and cultural inclusion and broader social justice [21], it is essential to evaluate how effectively the SNS addresses these barriers to healthcare for sanitation workers. However, measuring mere enrollment or nominal access to health insurance schemes can be misleading, as it does not capture the actual use of healthcare services or the barriers that prevent utilization. The frequency of utilization serves as a more dynamic and practical indicator of a scheme's effectiveness, reflecting not only the accessibility but also the acceptability and affordability of services [22]. Studies have shown that enrollment rates often overestimate real coverage, especially among marginalized groups, due to factors such as stigma, lack of awareness, administrative bottlenecks, and sociocultural constraints that limit service use [23, 24].

Therefore, analyzing how frequently beneficiaries use health services provides critical insights into both supply- and demand-side challenges and helps identify gaps that enrollment data alone cannot reveal. This study uniquely contributes to the literature as the first evaluation in Bangladesh to assess the frequency of SNS utilization among sanitation workers, examining how physical, socioeconomic, cultural and systemic factors interact to influence healthcare access. The findings provide critical insights for policymakers and stakeholders aiming to

enhance equitable healthcare delivery for this marginalized workforce.

Methods

Study area

The study was conducted across five municipalities in Bangladesh, Sakhipur, Saidpur, Lalmonirhat, Lakshmipur, and Teknaf, representing diverse urban, peri-urban, and regional contexts. Municipalities in the northern region (Sakhipur, Lalmonirhat, and Saidpur) provided insights into peri-urban and rural dynamics, where healthcare access and administrative resources are constrained. Lakshmipur Municipality in the southeast represented challenges typical of smaller, under-resourced local governments. Teknaf Municipality, located at the southernmost tip of the country, illustrated difficulties faced by remote, underdeveloped regions with limited healthcare facilities and administrative oversight. This selection facilitated a comprehensive understanding of sanitation workers across various geographical, administrative, and socioeconomic settings.

Sampling and data collection

Purposive sampling was employed to select sanitation workers enrolled in the Shasthya Nirapotta Scheme (SNS), targeting individuals directly relevant to the research objectives [25]. This non-probabilistic sampling approach was chosen due to the specific focus on SNS beneficiaries, limited sampling frames and logistical constraints in accessing this hard-to-reach population. A total of 155 sanitation workers enrolled in the SNS across the five municipalities participated in the survey. Although no formal sample size calculation was conducted, the sample size is comparable to that of similar studies evaluating healthcare utilization among marginalized workers [26]. Participants were recruited based on their enrollment status in SNS, with efforts to include diverse demographic and occupational profiles. Data collection was conducted through a structured questionnaire designed to capture demographic and socioeconomic characteristics, healthcare utilization patterns and potential barriers influencing scheme utilization, including social stigma, perceived scheme adequacy, premium affordability, geographic distance, travel time and administrative challenges.

Variable definition

The dependent variable in this study was healthcare utilization frequency under the Shasthya Nirapotta Scheme (SNS), representing the extent to which sanitation workers accessed healthcare services. Consistent with health service utilization theory, predicated on Andersen's behavioral model, which emphasizes that utilization reflects a range of predisposing, enabling,

and need-based factors [27, 28], this variable was categorized into three mutually exclusive, ordinal levels: Often, Sometimes and Hardly based on self-reported visit counts. If a worker experiences illness 'N' times in a year and reports utilizing SNS for treatment or care 'N' times, the utilization rate is calculated as $\frac{n}{N} \times 100\%$. The frequency of utilization was categorized as "Often" when the utilization rate $\geq 80\%$; "Sometimes" when $50\% \leq$ utilization rate $\leq 90\%$; and "Hardly" when the utilization rate $< 50\%$.

The "Hardly" category served as the reference group for regression modeling. Independent variables were selected on the basis of their theoretical relevance and empirical precedence, aligning with frameworks that link sociodemographic and occupational characteristics to healthcare utilization behavior [26, 28]. These variables included demographic characteristics as well as social and systemic factors affecting access to care.

Data analysis

Bivariate analysis was initially used to screen predictor variables using a liberal threshold of $p < 0.25$, following the purposeful selection strategy [29]. The dependent variable, healthcare utilization frequency under the Shasthya Nirapotta Scheme (SNS), is ordinal with three ordered categories: Often, Sometimes, and Hardly. Ordinal logistic regression was initially considered appropriate due to the natural ordering of the outcome. To validate this choice, the proportional odds assumption was assessed using the Brant Wald test, which examines whether predictor effects remain constant across all thresholds of the ordinal outcome [30, 31].

All statistical analyses were performed using IBM SPSS statistics (Version 27.0.1.0.) and R (Version 4.3.0). The Brant test for proportional odds was conducted in R using the brant package. Given that the Brant Wald test revealed a violation of the proportional odds assumption, multinomial logistic regression was chosen over ordinal regression and this decision is supported by recent evidence demonstrating its robustness in such contexts [32]. Unlike ordinal regression, this approach does not impose a proportionality constraint, enabling each category comparison to have unique coefficient estimates and thus more nuanced interpretations. Multinomial logistic regression is widely endorsed in applied social and health research when proportional odds assumptions are questionable, offering robustness and flexibility in modeling categorical outcomes without ordered constraints [33, 34]. This approach ensures valid inference and mitigates the risk of misleading conclusions associated with assumption violations in ordinal models.

Multinomial logistic regression models the log odds of each category of the dependent variable relative to a reference category [33, 35]. In this study, "Hardly" was selected

as the reference. The general form of the model used is as

$$\text{follows: } \ln \left[\frac{P(Y=j)}{P(Y=k)} \right] = \beta_{0j} + \beta_{1j}X_1 + \beta_{2j}X_2 + \dots + \beta_{kj}X_k$$

Where,

Y = dichotomous outcome

P (Y = j) is the probability of being in the outcome category j,

P (Y = k) is the probability of being in the reference category k,

$\ln \left[\frac{P(Y=j)}{P(Y=k)} \right]$ is the log (odds) of the outcomes,

β_{0j} is the intercept of category j

$\beta_{0j}, \beta_{1j}, \beta_{2j} \dots \beta_{kj}$ = Regression coefficients for predictor Xp in category j

$X_1, X_2, X_3 \dots$ = Predictor variables

The estimated coefficients were exponentiated to derive Adjusted Odds Ratios (AOR), and the results were presented with corresponding 95% Confidence Intervals (CIs). A p-value of <0.05 was considered statistically significant. To assess multicollinearity among independent variables, variance inflation factors (VIFs) were calculated, applying a conservative cutoff of VIF < 5 to ensure model stability, in line with established recommendations in epidemiological research [36].

Model adequacy was assessed using the Likelihood Ratio Test (LRT) based on the -2 Log-Likelihood (-2LL), which tests the null hypothesis that the inclusion of independent variables does not significantly improve model fit over the intercept-only model, as recommended in logistic regression methodology [33, 37]. The goodness-of-fit of the final model was further evaluated using

Pearson's chi-square and Deviance Chi-square statistics, where a p value greater than 0.05 indicates adequate fit [33, 35]. The predictive power of the model was assessed using Pseudo R² statistics, with higher values indicating a greater proportion of explained variation in the outcome.

Results

Socio-demographic characteristics

The sociodemographic characteristics of the respondents are shown in Table 1. The sanitation workforce in the selected study areas is predominantly male (60.6%), reflecting a gender dynamic in this occupational group. The majority of workers (79%) have been engaged in sanitation work for over five years, suggesting that they are well familiar with the challenges of the profession and capable of providing reliable and accurate information. While the average income of most workers (41%) is in the range of 150–299 BDT, a clear gender-based wage disparity persists as evidenced by the higher concentration of female workers in the lowest daily income bracket compared with their male counterparts. Most workers (74%) earn less than 300 BDT per day, whereas their monthly household expenditures often exceed 15,000 BDT (59%), reflecting a significant income-expenditure gap. Healthcare expenses are also substantial, with nearly half spending more than 3000 BDT per month. Monthly household expenditures were predominantly in the BDT 15,001–17,000 range (34.2%), while approximately one-third (29.0%) spent BDT 1000–2000 per month specifically on healthcare, suggesting financial prioritization of health within limited means. The majority worked 4–6 days per week (70.3%), indicating substantial work engagement

Table 1 Socio-demographic characteristics of the surveyed sanitation workers

Characteristics	Category	n (%) [Male (n), Female (n)]
Gender	Male	94 (60.6)
	Female	61 (39.4)
Average daily income (BDT)	100–149	50 (32.3) [20, 30]
	150–299	64 (41.3) [28, 37]
	300–449	19 (12.3) [16, 03]
	≥450	22 (14.2) [22, 00]
Average Monthly expenditure (BDT)	11000–13,000	33 (21.3) [16, 17]
	13001–15,000	30 (19.4) [10, 20]
	15001–17,000	53 (34.2) [17, 37]
	17001–19,000	39 (25.2) [17, 22]
Average Monthly expenditure for healthcare (BDT)	1000–2000	45 (29.0) [18, 27]
	2001–3000	39 (25.2) [14, 25]
	3001–4000	37 (23.9) [10, 27]
	>4000	34 (21.9) [15, 19]
Work experience as sanitation worker (Years)	0–5	33 (21.3) [15, 18]
	6–10	60 (38.7) [25, 36]
	11–15	39 (25.2) [13, 26]
	>15	23 (14.8) [15, 08]
Weekly working days	Daily	46 (29.7) [20, 26]
	1–3 days	00 (0) [00, 00]
	4–6 days	109 (70.3) [68, 41]

Table 2 Brant test of proportional odds assumption

Factors	Chi-square	df	Sig.
Global Omnibus Test	8.37	9	0.4
Gender	2.21	1	0.14
Average Daily Income	0.03	1	0.87
Average monthly expenditure for healthcare	0.17	1	0.68
Experience of social discrimination	4.25	1	0.6
Premium expense	0.28	1	0.04
Administrative bottlenecks	1.9	1	0.17
Facility distance	0.01	1	0.91
Travel Time	0.08	1	0.78
Perceived coverage inadequacy	0.31	1	0.45

Table 3 Model fitting information

Model	Model Fitting criteria	Likelihood Ratio Test		
	-2 Log Likelihood	Chi-Square	df	Sig.
Intercept Only	324.495	—	—	—
Final	229.877	94.619	26	< 0.001

Table 4 Pseudo R-Square

Statistic	Value
Cox and Snell	0.457
Nagelkerke	0.515
McFadden	0.279

that could impact both their time availability and physical need for accessing healthcare.

Statistic modeling and model fit

The result from Brant Wald test (Table 2) reveals that while the overall test was non-significant ($p = 0.40$), indicating partial model adequacy, the predictor “Premium Expense” significantly violated this assumption ($p = 0.04$). Such violations, even if limited to a single key predictor, may produce biased and inconsistent estimates under an ordinal logistic model [37, 39]. Given the substantive role of “Premium Expense” and the violation of proportional odds, a multinomial logistic regression model was adopted. Multicollinearity among the independent variables was evaluated using Variance

Inflation Factors (VIF), with all values below 2, indicating no significant multicollinearity and ensuring stable parameter estimates. Variables meeting p -value threshold of < 0.25 criterion—gender, average daily income, average monthly healthcare expenditure, social discrimination, premium expenses, administrative bottlenecks, facility distance, travel time, and perceived coverage inadequacy were subsequently included in the final multinomial logistic regression model. Model adequacy was evaluated via the Likelihood Ratio Test (LRT), comparing the fitted model to an intercept-only baseline. The significant LRT statistic ($\chi^2 = 94.619$, $df = 26$, $p < 0.001$) and a reduction in the -2 Log-Likelihood from 324.495 to 229.877, shown in Table 3, indicate that the inclusion

of predictors substantially improves model fit. Goodness-of-fit was further assessed using Pearson’s Chi-square and Deviance Chi-square tests, with non-significant results ($p > 0.05$) supporting an adequate fit between the model and the data.

The model’s explanatory power was quantified using Pseudo R^2 statistics (Table 4), including Cox and Snell (0.457), Nagelkerke (0.515), and McFadden (0.279). Although logistic regression lacks a direct equivalent to the R^2 used in linear regression, the Pseudo R^2 values, supported by significant likelihood-ratio and goodness-of-fit statistics indicate that the model explains a moderate to substantial proportion (approximately 28% to 52%) of the variability in healthcare utilization frequency. Such values fall well within the range regarded as a strong model fit in social-behavioral research [40].

Factors affecting the effectiveness of SNS

Multinomial logistic regression revealed (Table 5) that gender, social discrimination, premium expenses, administrative bottlenecks, facility distance, and travel time were significant determinants of healthcare utilization frequency under the Shasthya Nirapotta Scheme (SNS).

Males had 72% lower odds of using SNS “sometimes” compared to females (AOR = 0.28, 95% CI: 0.09–0.82, $p = 0.021$). Workers reporting no social discrimination were eight times more likely to use SNS “often” compared to those experiencing discrimination (AOR = 8.36, 95% CI: 1.98–35.15, $p = 0.004$). Perception of the SNS premium as “not expensive” increased the odds of “often” utilization compared to those perceiving it as “expensive” (AOR = 13.75, 95% CI: 3.73–50.68, $p < 0.001$). The absence of administrative bottlenecks increased the odds of frequent SNS use by 24 times (AOR = 24.21, 95% CI: 5.96–98.27, $p < 0.001$).

Perceiving the healthcare facility as conveniently located was associated with higher odds of frequent utilization (AOR = 15.57, 95% CI: 3.87–62.69, $p < 0.001$), while lack of travel time burden also increased the odds (AOR = 9.61, 95% CI: 2.57–35.88, $p < 0.001$).

Discussion

This study examined key determinants influencing the frequency of healthcare utilization among sanitation workers, a vital indicator of the Shasthya Nirapotta Scheme’s (SNS) effectiveness. Findings reveal that physical, socio-economic, cultural, and institutional barriers constrain optimal use. Significant factors include gender, social discrimination, premium expenses, administrative bottlenecks, facility distance, and travel time.

Sanitation workers in South Asia experience entrenched sociocultural exclusion rooted in caste, class, and occupational hierarchies, systematically marginalizing them from essential social rights, including

Table 5 Factors affecting the effectiveness of SNS

Factors		Frequency of utilization of SNS			
		Comparison between “Often” and “hardly” utilization		Comparison between “Sometimes” and “hardly” utilization	
		AOR (95% CI)	P value	AOR (95% CI)	P value
Gender	Female	Ref			
	Male	0.45 (0.13 — 1.58)	0.216	0.28 (0.09 — 0.82)	0.021**
Average daily income	≥450	Ref	0.149	1.09 (0.19 — 6.39)	0.799
	300–449	4.90 (0.56 — 42.51)	0.607	1.31 (0.29 — 5.89)	0.729
	150–299	1.62 (0.25 — 10.35)	0.707	0.76 (0.12 — 5.07)	0.925
	100–149	1.55 (0.16 — 15.06)			
Average monthly expenditure for healthcare	> 4000	Ref	0.127	2.00 (0.39 — 10.38)	0.275
	3001–4000	0.276 (0.05 — 1.44)	0.362	0.54 (0.14 — 2.11)	0.379
	2001–3000	0.464 (0.09 — 2.42)	0.129	0.48 (0.126 — 1.80)	0.408
	1000–2000	4.222 (0.66 — 27.12)			
Experience of social discrimination	No	8.359 (1.98 — 35.15)	0.004**	2.31 (0.70 — 7.62)	0.167
	Yes	Ref			
Premium expense	Not expensive	13.754 (3.73 — 50.68)	< 0.001**	1.30 (0.43 — 3.92)	0.647
	Expensive	Ref			
Administrative bottlenecks	No	24.209 (5.96 — 98.23)	< 0.001**	3.28 (1.08 — 9.97)	0.036*
	Yes	Ref			
Facility distance	Disagree/No	15.573 (3.87 — 62.68)	< 0.001**	4.45 (1.40 — 14.22)	0.012**
	Agree/yes	Ref			
Travel Time	No	9.608 (2.57 — 35.89)	< 0.001**	2.14 (0.74 — 6.14)	0.158
	Yes	Ref			
Perceived coverage inadequacy	Agree	Ref	0.389	1.27 (0.49 — 3.25)	0.623
	Disagree	0.612 (0.20 — 1.87)			

* $p < 0.05$, ** $p < 0.025$

healthcare access [38, 41]. Social exclusion is a dynamic, multidimensional process driven by unequal power relations across economic, political, social, and cultural domains [21]. The stigmatization and discrimination associated with sanitation work foster mistrust and feelings of unwelcome, deterring many workers from seeking medical care [42]. Our findings corroborate that barriers to healthcare extend beyond structural availability and are embedded within broader social processes where social norms, cultural attitudes, and interpersonal discrimination intersect to shape health-seeking behavior.

Economic vulnerability emerged as another critical determinant. Low wages and a substantial income-expenditure gap render premium expenses decisive in healthcare utilization. Evidence from India demonstrates that out-of-pocket expenditures and premium-based schemes often exclude informal and low-wage workers due to affordability issues, even when subsidies exist [43]. Our study further reveals perceptual differences in affordability among workers with similar incomes: some viewed premiums as affordable, whereas others did not. This suggests that affordability transcends economics, reflecting how individuals prioritize healthcare amid competing needs. Hence, addressing financial barriers alone is insufficient; shifting social attitudes to recognize healthcare as an indispensable right is equally important [44, 45].

Administrative efficiency critically influences service uptake. Systemic inefficiencies—delays, misinformation, and service denials—discourage care seeking [22]. For informal workers lacking formal documentation, these hurdles are amplified. Additionally, healthcare providers’ implicit and explicit biases against marginalized groups, including sanitation workers predominantly from Dalit communities, exacerbate access barriers through neglect or abuse [46]. Our findings affirm that administrative bottlenecks and discrimination significantly impede healthcare scheme success among marginalized populations.

Geographical marginalization, reflected in facility distance and travel time, further limits access. Respondents traveling long distances were significantly less likely to utilize SNS services, which is consistent with prior research identifying geographic accessibility as a fundamental determinant in low- and middle-income contexts [47, 48]. These challenges are intensified by demanding inflexible work schedules common in informal sectors, limiting discretionary time for healthcare [49, 50]. Moreover, sanitation workers’ spatial segregation into informal peri-urban settlements excludes them from urban planning and service delivery, reinforcing systemic inequalities [51]. Therefore, incorporating a location-sensitive healthcare design is essential for enhancing scheme utilization.

Conclusions

This study demonstrated that the effectiveness of health-care schemes such as the Shasthya Nirapotta Scheme (SNS) for sanitation workers cannot be fully understood through socioeconomic factors alone. While economic vulnerability—including low wages and out-of-pocket costs—remains a significant barrier, utilization is equally shaped by sociocultural, institutional, and spatial determinants. Perceived affordability, social stigma, administrative inefficiencies, geographic isolation, and time poverty interact to create a complex exclusionary landscape that restricts healthcare access for this marginalized group.

These findings underscore the need for a multidimensional approach that integrates economic support with targeted social and institutional reforms. Enhancing healthcare access for sanitation workers requires deliberate efforts to combat stigma, promote administrative inclusivity, and design services that are geographically and culturally appropriate. Public health policies must explicitly address sociocultural determinants of exclusion, recognizing that marginalized communities require more than infrastructure alone. Equity-driven interventions prioritizing dignity, respect, and institutional inclusion are essential for healthcare schemes like SNS to achieve their goals and meaningfully reduce health inequities in Bangladesh.

Abbreviations

AOR	Adjusted Odds Ratios
CI	Confidence Interval
LGIs	Local government institutions
MDG	Millennium Development Goal
SDG	Sustainable Development Goal
SNS	Shasthya Nirapotta Scheme
SRMS	Self-Employment Scheme for Rehabilitation of Manual Scavengers
VIFs	Variance inflation factors

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Authors contributions

F.T. and M.Z.O. conceived and designed the study, conducted literature review, developed the methodology, analyzed and interpreted data, drafted the initial manuscript and revised based on review. N.U.S. conceptualized and refined the methodology, conducted literature review, performed data analysis, critically reviewed, drafted and revised the manuscript. M.T.I. and H.J. contributed to study design and critically reviewed the manuscript. H.I.A. and M.M.M. contributed data collection and critically reviewed the manuscript. F.T. and M.Z.O. prepared the first draft, while all authors critically reviewed, revised and approved the final version of the manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study involved obtaining informed consent, ensuring confidentiality, and no risks to participants. It adheres to principles of fairness, integrity, and transparency, with ethical approval from the WaterAid Ethical Review Board. Cultural sensitivity and respect for participant autonomy were prioritized.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Civil Engineering, Bangladesh University of Engineering and Technology, Palashi, Dhaka 1000, Bangladesh

²WaterAid Bangladesh, House 97/B, Road 25, Block A, Banani, Dhaka 1213, Bangladesh

³Institute of Statistical Research & Training, University of Dhaka, Secretariate Road, Dhaka 1000, Bangladesh

⁴BRAC International, BRAC Centre, 75 Mohakhali, Dhaka 1212, Bangladesh

⁵Save the Children, House No. CWN (A) 35, Road No. 43, Gulshan 2, Dhaka 1212, Bangladesh

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